E2E Final Board

DB1

Customer	Test Customer
Company profile	iec-default
Design rules	software defaults, not confirmed by the company's engineers

Rev.	Date	Description	Drawn	Checked	Review approved
01	2026-10-23	First issue			
02	2026-10-23	AC raised to 3.5 kW			

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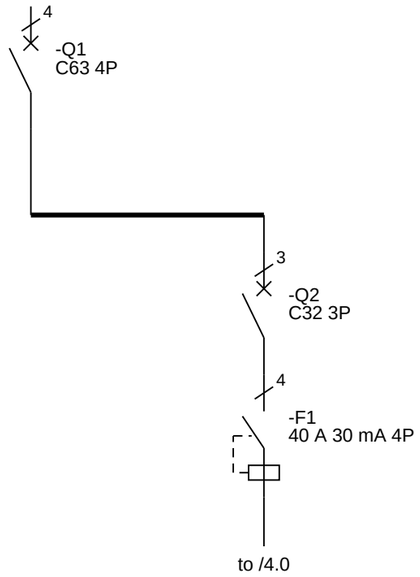
3/N/PE ~ 50 Hz
400 V TN-S



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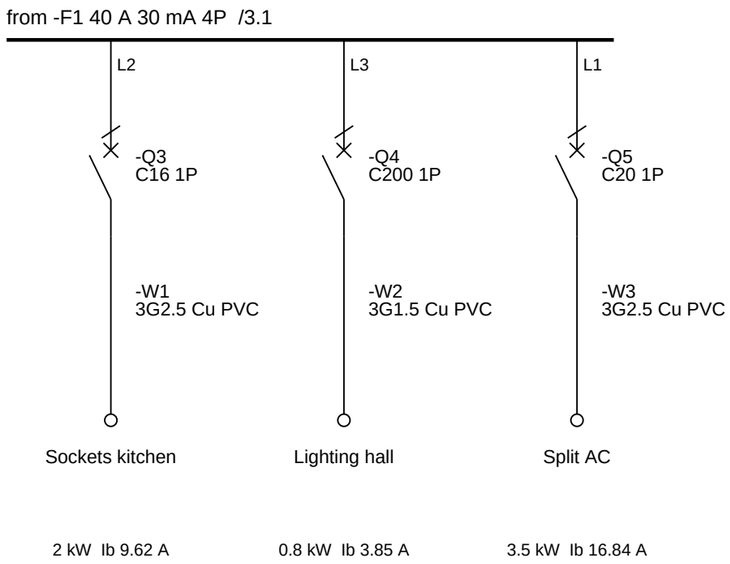

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No.	Note
1	Split AC: Ib 16.84 A exceeds the company's 16 A for air_conditioning; rated 20 A instead.
2	Group 1 breaker raised to 32 A, 1.6 x the 20 A breaker after it, for discrimination.
3	Phase imbalance 77.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
4	Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
5	No fault level given: every breaker's breaking capacity is left to be confirmed.
6	Leave 1 spare outgoing ways (20 %).
7	Loads without a power factor were taken at cos phi 0.9.
8	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
9	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
10	Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings.
11	Circuit rules are this software's defaults, not confirmed by the company's engineers.
12	Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 18 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

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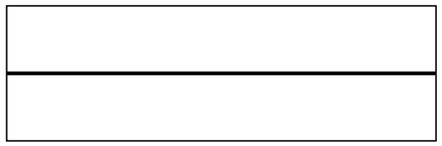
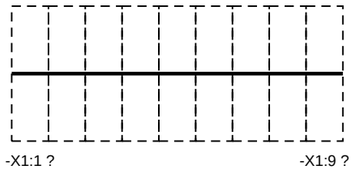
Incomer: 70 mm of rail



Residual current group: 157.5 mm of rail; 1 without a width

Terminals: 0 mm of rail; 9 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

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Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L2	Sockets kitchen	-W1	2.5	8WH1000-0AF00
-X1:2	N	Sockets kitchen	-W1	2.5	8WH1000-0AF00
-X1:3	PE	Sockets kitchen	-W1	2.5	8WH1000-0CF07
-X1:4	L3	Lighting hall	-W2	1.5	not selected
-X1:5	N	Lighting hall	-W2	1.5	not selected
-X1:6	PE	Lighting hall	-W2	1.5	not selected
-X1:7	L1	Split AC	-W3	2.5	8WH1000-0AF00
-X1:8	N	Split AC	-W3	2.5	8WH1000-0AF00
-X1:9	PE	Split AC	-W3	2.5	8WH1000-0CF07

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Cable	From	To	Type	Length
DB1 -W1	DB1 -Q3	Sockets kitchen	3G2.5 Cu PVC	20 m
DB1 -W2	DB1 -Q4	Lighting hall	3G1.5 Cu PVC	20 m
DB1 -W3	DB1 -Q5	Split AC	3G2.5 Cu PVC	20 m

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Qty	Description	Article	Order number	Devices
1	Circuit-breaker C63 4P	not selected		-Q1
1	Circuit-breaker C32 3P	not selected		-Q2
1	Residual current circuit-breaker 40 A 30 mA 4P	not selected		-F1
1	Circuit-breaker C16 1P	not selected		-Q3
1	Circuit-breaker C200 1P	not selected		-Q4
1	Circuit-breaker C20 1P	not selected		-Q5
4	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	DB1 -X1
2	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	DB1 -X1

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Findings (1 blocking, 5 warning, 7 info), page 1 of 1

[BLOCKING] DB1 / c2: Lighting hall: 200 A protection is above the 17.5 A its 3G1.5 cable carries (method B1); In <= Iz does not hold. (integrity_device_above_cable_capacity; source: ABB Electrical installation handbook Vol. 2, §2.3 (In <= Iz); Iz from §2.2.1 Tables 4, 5, 8)

[WARNING] DB1 / c3: Split AC: Ib 16.84 A exceeds the company's 16 A for air_conditioning; rated 20 A instead. (above_company_rating; source: policy; engineering confirmation required)

[WARNING] DB1: Phase imbalance 77.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given. (phase_imbalance; source: company max_phase_imbalance_percent; engineering confirmation required)

[WARNING] DB1: No fault level given: every breaker's breaking capacity is left to be confirmed. (no_fault_level; source: policy; engineering confirmation required)

[WARNING] DB1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)

[WARNING] DB1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)

[INFO] DB1: Group 1 breaker raised to 32 A, 1.6 x the 20 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)

[INFO] DB1: Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)

[INFO] DB1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)

[INFO] DB1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)

[INFO] DB1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)

[INFO] DB1: Designed for Palestine, under IEC 60364 (adopted by the Palestine Standards Institution): its stated limits apply beneath the company's own settings. (market_rules; source: policy)

[INFO] DB1: Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 18 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted. (enclosure_selected; source: policy)